

1. Creating and Renaming Files/Directories

Create a directory named test_dir using mkdir.

Inside test_dir, create an empty file called example.txt.

Rename example.txt to renamed_example.txt using mv

```
root@ubuntu:~$ mkdir test_dir
root@ubuntu:~$ ls
filesystem  test_dir
root@ubuntu:~$ cd test_dir/
root@ubuntu:~/test_dir$ mkdir example.txt
root@ubuntu:~/test_dir$ ls
example.txt
root@ubuntu:~/test_dir$ mv example.txt/ renamed_example.txt
root@ubuntu:~/test_dir$ ls
renamed_example.txt
root@ubuntu:~/test_dir$ █
```

Ans:

2. Viewing File Contents

Use cat to display the contents of /etc/passwd.

Display only the first 5 lines of /etc/passwd using head.

Display only the last 5 lines of /etc/passwd using tail.

```

root@ubuntu:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uidd:x:103:103::/run/uidd:/usr/sbin/nologin
tss:x:104:104:TPM software stack,,,:/var/lib/tpm:/bin/false
sshd:x:105:65534::/run/sshd:/usr/sbin/nologin
pollinate:x:106:1::/var/cache/pollinate:/bin/false
tcpdump:x:107:108::/nonexistent:/usr/sbin/nologin
landscape:x:108:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:990:990:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
polkitd:x:989:989:User for polkitd:/:/usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
kc-internal:x:0:0:/:root:/bin/bash
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
root@ubuntu:~$ █

```

Ans:

```

root@ubuntu:~$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
root@ubuntu:~$ tail -n 5 /etc/passwd
fwupd-refresh:x:990:990:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
polkitd:x:989:989:User for polkitd:/:/usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
kc-internal:x:0:0:/:root:/bin/bash
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
root@ubuntu:~$ █

```

3. Searching for Patterns

Use grep to find all lines containing the word "root" in /etc/passwd.

```
root@ubuntu:~$ grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
kc-internal:x:0:0:/:root:/bin/bash
root@ubuntu:~$
```

Ans:

4. Zipping and Unzipping

Compress the test_dir directory into a file named test_dir.zip using zip.

Unzip test_dir.zip into a new directory named unzipped_dir.

Ans:

```
root@ubuntu:~$ zip -r test_dir.zip test_dir
Command 'zip' not found, but can be installed with:
apt install zip
root@ubuntu:~$ apt install zip
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  squashfs-tools
Use 'apt autoremove' to remove it.
The following NEW packages will be installed:
  zip
0 upgraded, 1 newly installed, 0 to remove and 53 not upgraded.
Need to get 176 kB of archives.
After this operation, 549 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 zip amd64 3.0-13ubuntu0.2 [176 kB]
Fetched 176 kB in 0s (4627 kB/s)
Selecting previously unselected package zip.
(Reading database ... 114047 files and directories currently installed.)
Preparing to unpack .../zip_3.0-13ubuntu0.2_amd64.deb ...
Unpacking zip (3.0-13ubuntu0.2) ...
Setting up zip (3.0-13ubuntu0.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ubuntu:~$ zip -r test_dir.zip test_dir
  adding: test_dir/ (stored 0%)
  adding: test_dir/renamed_example.txt/ (stored 0%)
root@ubuntu:~$
```

```

root@ubuntu:~$ mkdir unzipped_dir
root@ubuntu:~$ ls
filesystem test_dir test_dir.zip unzipped_dir
root@ubuntu:~$ unzip test_dir.zip -d unzipped_dir
Archive: test_dir.zip
  creating: unzipped_dir/test_dir/
  creating: unzipped_dir/test_dir/renamed_example.txt/
root@ubuntu:~$ ls
filesystem test_dir test_dir.zip unzipped_dir
root@ubuntu:~$ cd unzipped_dir/
root@ubuntu:~/unzipped_dir$ ls
test_dir
root@ubuntu:~/unzipped_dir$ █

```

5. Downloading Files

Use wget to download a file from a URL (e.g., <https://example.com/sample.txt>).

Ans:

```

root@ubuntu:~$ wget https://raw.githubusercontent.com/github/gitignore/main/README.md
--2026-06-17 12:02:29-- https://raw.githubusercontent.com/github/gitignore/main/README.md
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.110.133, 185.199.111.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 5624 (5.5K) [text/plain]
Saving to: 'README.md'

README.md          100%[=====>]  5.49K  --.-KB/s  in 0s

2026-06-17 12:02:30 (101 MB/s) - 'README.md' saved [5624/5624]

root@ubuntu:~$ ls
README.md filesystem test_dir test_dir.zip unzipped_dir

```

6. Changing Permissions

Create a file named secure.txt and change its permissions to read-only for everyone using chmod.

```

root@ubuntu:~$ chmod 444 secure.txt
root@ubuntu:~$ ls
README.md filesystem secure.txt test_dir test_dir.zip unzipped_dir
root@ubuntu:~$ ls -l
total 20
-rw-r--r-- 1 root root 5624 Jun 17 12:02 README.md
lrwxrwxrwx 1 root root    1 May 16 19:59 filesystem -> /
-r--r--r-- 1 root root    0 Jun 17 12:03 secure.txt
drwxr-xr-x 3 root root 4096 Jun 17 11:51 test_dir
-rw-r--r-- 1 root root 354 Jun 17 11:57 test_dir.zip
drwxr-xr-x 3 root root 4096 Jun 17 11:59 unzipped_dir
root@ubuntu:~$ █

```

Ans:

7. Working with Environment Variables

Use export to set a new environment variable called MY_VAR with the value "Hello, Linux!".

Ans:

```
root@ubuntu:~$ export MY_VAR="Hello, Linux!"
root@ubuntu:~$ echo $MY_VAR
Hello, Linux!
root@ubuntu:~$ █
```

Github Repo Link: <https://github.com/naganesandip07-sketch/Devops>